

**The nursing process for a patient with postoperative septic shock following a
laparotomy in intensive care**

**Proceso de enfermería a persona con choque séptico postoperada de laparotomía en
terapia intensiva**

**Processo de enfermagem à pessoa com choque séptico após cirurgia de
laparotomia em terapia intensiva**

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Abstract

Introduction: Septic shock is a life-threatening multisystem disorder caused by a severe infection that compromises tissue perfusion. In this context, applying the nursing process is essential in order to provide comprehensive, personalized, evidence-based care. **Objective:** To apply the nursing process to a patient with septic shock of intestinal origin, who has undergone exploratory laparotomy and is hospitalized in the ICU. **Methodology:** A clinical case study of a 41-year-old female patient was conducted based on the nursing process and Gordon's Functional Health Patterns. Standardized language was used for the nursing diagnoses, expected outcomes and nursing interventions. An assessment was performed to identify altered human needs and establish care plan priorities. Ethical considerations were taken into account, with personal data protected through anonymity and confidentiality. **Results:** Six priority nursing diagnoses were identified, namely, risk of shock; impaired gas exchange; ineffective airway clearance; risk of bronchoaspiration; hypothermia; and impaired tissue integrity. Expected outcomes included hemodynamic stabilization, improved respiratory status and progression in wound healing. Proposed nursing interventions included managing septic shock, assisting with mechanical ventilation, suctioning secretions, and promoting wound healing, all of which contributed to the patient's improvement. **Conclusions:** Applying the nursing process enabled systematic care to be provided, critical thinking to be used, and standardized nursing language to be employed. This promoted care that focused on the individual needs of critically ill patients with septic shock.

Key words: Nursing process; Septic shock; Critical care nursing (DeCS).

Resumen

Introducción: El estado de choque séptico es una alteración multisistémica secundaria a una infección grave, que compromete la perfusión tisular y pone en riesgo la vida, en este contexto la aplicación del proceso de enfermería es indispensable para proporcionar cuidados integrales, individualizados y basados en la evidencia. **Objetivo:** Aplicar el proceso de enfermería a una persona en estado de choque séptico de origen intestinal posterior a laparotomía exploradora hospitalizada en terapia intensiva. **Metodología:** Estudio de caso clínico a femenina de 41 años, fundamentado en el Proceso de Enfermería basado en Patrones Funcionales de Salud de Gordon, utilizando lenguaje estandarizado de Diagnósticos de Enfermería, Resultados Esperados e Intervenciones de Enfermería. Se realizó valoración identificando necesidades humanas alteradas, estableciendo prioridades en plan de cuidados, como consideraciones éticas se resguardaron los datos personales mediante anonimato y confidencialidad. **Resultados:** Se identificaron seis diagnósticos de enfermería prioritarios: riesgo de shock, deterioro del intercambio gaseoso, limpieza ineficaz de vías aéreas, riesgo de aspiración, hipotermia y deterioro de la integridad tisular. Los resultados esperados incluyeron estabilización hemodinámica, mejora del estado respiratorio y progresión en la cicatrización de heridas. Las intervenciones de enfermería planteadas incluyeron manejo del shock séptico, asistencia en la ventilación mecánica, aspiración



de secreciones y curación de heridas, contribuyendo a la mejora de la paciente. **Conclusiones:** La aplicación del proceso de enfermería permitió brindar cuidados sistematizados, emplear el pensamiento crítico y utilizar el lenguaje estandarizado de enfermería, favoreciendo una atención centrada en las necesidades individuales de la persona en estado crítico con choque séptico.

Palabras clave: Proceso de enfermería; Choque séptico; Enfermería de cuidados críticos (DeCS).

Abstrato

Introdução: Choque séptico refere-se a alterações multissistêmicas secundárias a uma infecção grave, que comprometem a perfusão tecidual e são potencialmente fatais. Nesse contexto, a aplicação do processo de enfermagem é essencial para proporcionar um cuidado integral, individualizado e baseado em evidências. **Objetivo:** Aplicar o processo de enfermagem a uma paciente em choque séptico de origem intestinal após laparotomia exploradora, internada em terapia intensiva. **Metodologia:** Estudo de caso clínico de uma paciente do sexo feminino, 41 anos, com base no processo de enfermagem baseado nos padrões funcionais de saúde de Marjory Gordon, utilizando diagnósticos de enfermagem padronizados, resultados esperados e intervenções de enfermagem. Foi realizada uma avaliação, identificando necessidades humanas alteradas e estabelecendo prioridades no plano de cuidados. Como considerações éticas, os dados pessoais foram protegidos por meio do anonimato e da confidencialidade. **Resultados:** Foram identificadas intervenções de enfermagem para seis diagnósticos de enfermagem prioritários, nomeadamente, risco de choque, troca gasosa prejudicada, desobstrução ineficaz das vias aéreas, risco de aspiração, hipotermia e integridade tecidual prejudicada. Os resultados esperados incluíram estabilização hemodinâmica, melhora do estado respiratório e cicatrização progressiva da ferida. As intervenções incluíram manejo do choque séptico, assistência ventilatória mecânica, aspiração de secreções e cuidados com feridas. **Conclusões:** A aplicação do processo de enfermagem possibilitou a prestação de cuidados sistemáticos, o uso do pensamento crítico e a utilização de uma linguagem de enfermagem padronizada. Isso promoveu um cuidado focado nas necessidades individuais de pacientes em estado crítico com choque séptico.

Palavras-chave: Processo de enfermagem; Choque séptico; Enfermagem em terapia intensiva (DeCS).



Introduction

The nursing profession involves providing care based on specific knowledge and techniques for individuals, families, or communities, using the Nursing Process (NP) methodology. This methodology is defined as a set of logical, dynamic, and systematic steps aimed at providing nursing care based on scientific evidence and critical thinking ⁽¹⁾.

During the nursing process, nurses assess and diagnose patients, planning, executing and evaluating interventions aimed at improving and/or maintaining health. They consider the totality, safety and continuity of care required ⁽²⁾. Critically ill patients are usually treated in intensive care units (ICUs) or critical care units. One of the most complex problems in this context is the septic process, which is a major cause of morbidity and mortality, particularly when multiple organ dysfunctions are present. This affects different organs and systems, gradually progressing to septic shock. This results in prolonged ICU stays, high hospital costs, and a decrease in the quality of life of surviving patients ⁽³⁾.

The severity of patients' illnesses and complications when they are admitted to the ICU makes it essential to provide them with the necessary care and continuity of care. This emphasizes the importance of implementing the NP as a scientific methodology in the ICU, thereby ensuring high-quality care that considers the various needs of patients ^(4,5). Marjory Gordon's Functional Health Patterns (FHP) assessment framework provides a comprehensive overview of how individuals, families and communities respond to various life processes and/or health issues. The framework is based on eleven FHPs, which enable a holistic patient assessment and the accurate identification of human responses. This allows for the selection of appropriate nursing interventions to improve life processes, resolve health problems, and reduce risks ⁽⁶⁾.

Septic shock is a challenging pathology for the healthcare sector, nursing staff and, of course, those who experience it and their families. In the case of public hospitals, the healthcare sector faces high

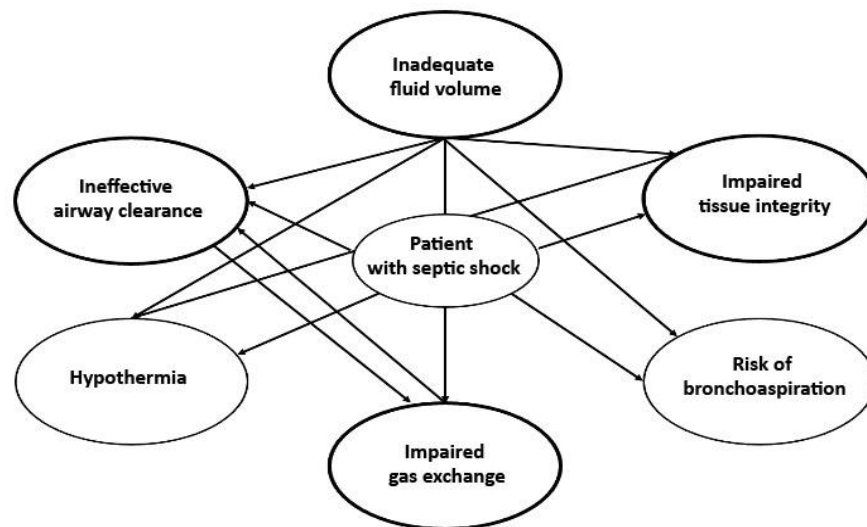


economic costs associated with complications and procedures, as well as increased mortality rates. For nursing staff, it is challenging due to the complexity of the care involved. For patients and their families, it is challenging because their lives are at risk due to the complex pathology induced by the host's immune system response to an infectious process ⁽⁷⁾. Therefore, we present a clinical case in which we apply the NP to a patient with intestinal septic shock following exploratory laparotomy admitted to the ICU.

Methodology

This is a single clinical case study of a 41-year-old woman in the ICU over the course of three days of nursing intervention. It is based on the 11 PFS nursing process for nursing assessment, developed by Marjory Gordon. The NP was developed in accordance with the five stages: A nursing assessment was performed, during which data was collected through physical examination and secondary sources. A critical analysis was then carried out, documenting the findings and identifying six nursing diagnoses, which were prioritized using the Analysis of Current State Results (in Spanish Area) nursing model as a clinical reasoning tool to analyze the results of the current status, next, care was planned, (Figure 1).

Figure 1. Nursing diagnosis reasoning network according to the AREA model, 2025.



Source: Self-developed



In the planning stage, care plans were developed using the professional taxonomies of Nursing Diagnoses (NANDA-I), Expected Outcomes (NOC) and Nursing Interventions (NIC) to facilitate the implementation and evaluation of interventions by nursing staff, and to assess their effectiveness and efficiency. The information obtained is documented in the care plan formats and details the interventions performed and the patient's progress. The study was based on ethical principles that guaranteed anonymity and confidentiality during the collection of data, as well as respect for the following ethical principles relating to patients: autonomy, justice, non-maleficence and beneficence ⁽¹²⁾.

Case presentation:

M.C.A.G., is a 41-year-old Mexican female patient who was brought by ambulance to the emergency department of a public hospital with suspected pneumonia due to bronchoaspiration and infection of a surgical wound following exploratory laparotomy secondary to intestinal obstruction with intestinal diversion (ascending colostomy). Family history includes systemic arterial hypertension and hypercholesterolemia on both paternal and maternal sides.

The patient was admitted on february 15, 2023, in a drowsy state with signs of respiratory distress: Tachypnoea (30 breaths per minute), intercostal retractions and respiratory grunting. She was therefore given oxygen therapy with a reservoir mask at a flow rate of 10 liters per minute. She presented with sinus tachycardia (120 beats per minute), arterial hypotension (80/40 mmHg), hypoxaemia (85 %), and signs of tissue hypoperfusion (generalized pallor and a capillary refill time of six seconds). Fluid therapy was initiated with Hartmann's solution at 125 ml/h and norepinephrine at 5 ml/h. Intestinal hypoperistalsis and an absence of intestinal elimination via the ostomy were identified. Urinary elimination was achieved via a Foley catheter, revealing oliguria (0.3 ml/kg/h) and high urine concentration.



Subsequently, she continued to show signs of septic shock and hemodynamic instability, necessitating a progressive increase in the vasopressor dose to 15 ml/h; additionally, there was an increase in respiratory failure, with hypoxaemia (83 %) and tachypnoea at 35 breaths per minute (bpm), despite oxygen therapy support with a reservoir mask. Invasive respiratory support was indicated, so orotracheal intubation was performed using an 8-gauge tube and assisted-controlled mechanical ventilation with an inspired oxygen fraction (FiO_2) of 100 %, an inspiratory pressure (IP) of 15 cmH_2O , a positive end-expiratory pressure (PEEP) of 7 cmH_2O , an inspiration-to-expiration ratio (I:E) of 1:2 and a cycle frequency (CF) of 15 per minute. Support was provided through deep sedation and analgesia using an infusion of midazolam (100 mg/100 ml of 0.9% sodium chloride solution) at 14 ml/h, undiluted propofol (1 g/100 ml) at 15 ml/h and undiluted fentanyl (500 μg /10 ml) at 2 ml/h, onium 250 mg diluted in 100 ml of 0.9 % sodium chloride solution at 7 ml/h. Subsequently, antibiotic therapy with carbapenems and lincosamides was initiated.

Secondary sources reported mixed acidosis, leukocytosis and thrombocytopenia. A chest X-ray showed pulmonary radiolucency in the right apex and bilateral alveolar infiltration. Two days later, she was admitted to the ICU due to septic shock following exploratory laparotomy for intestinal obstruction, with subsequent intestinal diversion (ascending colostomy). She also presented with acute respiratory distress syndrome resulting from bronchoaspiration pneumonia and right pneumothorax.

The nursing assessment was performed three days after admission. With regard to nutritional and metabolic status, total parenteral nutrition was provided at an intake level of 1400 kcal, alongside fasting until further notice. The skin was intact with slight generalized pallor, adequate turgor and normal temperature in all areas. According to Kosier, axillary temperature measurement showed hypothermia (35.5 °C). The body mass index was 43.3 kg/m^2 , indicating type III obesity according



to NOM-008-SSA3-2017. A nasogastric tube was inserted to bypass the biliary system. According to the Godette scale, he had grade 2 edema in the upper and lower limbs (++). According to the Seidel Guide to Physical Examination, the head, scalp, face, neck and oral cavity showed no abnormalities.

The patient had a wound in the hypogastrium due to an exploratory laparotomy. There were sutures in the proximal two-thirds and slight visualization of granulation tissue in the distal third. There were also peripheral erythema and redness, as well as exposed passive drainage (Penrose) with purulent, slightly foul-smelling contents. This was correctly implanted. She also had a stage 2 pressure injury in the sacral area. The 24-hour fluid balance was positive at 676.5 ml. Laboratory tests showed hyperglycemia (153 mg/dL), leukocytosis ($14.72 \times 10^3/\mu\text{L}$), lymphopenia (47%), neutrophilia (91.9%) and normocytic hypochromic anemia (hemoglobin: 8.3 g/dL, mean corpuscular volume: 87 fL and mean corpuscular hemoglobin: 28.9 g/dL).

In terms of activity-exercise patterns, a normothorax was observed, with normal proportions between the anteroposterior and transverse diameters. The patient was undergoing assisted mechanical ventilation with established dynamic parameters: Assist/Control mode; pressure mode; IP: 15 cmH₂O; FiO₂: 50 %, PEEP: 8 cmH₂O and a heart rate of 18 per minute. An active humidifier was also in use. The patient had rales in the tracheal area. Oxygen saturation was borderline (93%), but within the normal range according to PISA. Tachypnoea (35 rpm) was evident, confirming moderate decoupling from the mechanical ventilator. Rales were present in the apices and bases of the lung fields. The patient had mild arterial hypotension (109/80 mmHg) and a mean arterial pressure (MAP) of 89 mmHg. She was dependent on a continuous infusion of 16 mg of norepinephrine in 100 ml of 5 % glucose solution at 18 ml/h. According to Seidel's physical examination guidelines, she had a response dose, sinus tachycardia (110 beats per minute) and normal central heart sounds.



The patient had a three-lumen FR 7 central venous catheter inserted in the right subclavian vein on 16 February 2023. There were no signs of bleeding or secretions. She had a prolonged capillary refill time of six seconds. She is at high risk of falling, with a Downton scale score of 5. Secondary sources revealed mixed acidosis (pH: 7.23; pCO₂: 82 mmHg; PO₂: 155 mmHg; HCO₃ 34.3 mmol/L; EB: -9). An anteroposterior chest X-ray showed diffuse alveolar infiltration affecting the airspace, as well as bilateral opacities at the bases and radiolucency at the right apex.

No alterations were observed in the sensory organs regarding cognitive-perceptual patterns, except for pupillary hyporeactivity to light, according to the Seidel Guide to Physical Examination. On the RASS scale, he scored -2, indicating moderate sedation. He showed slight eye opening and slight movement of the lower limbs but did not overcome gravity. Using the Behavioral Pain Scale (BPS), he was classified as experiencing unacceptable pain (6 points), demonstrating disconnection from the ventilator and movement of the lower limbs and slight facial contraction.

No alterations were observed in the genitals or mammary glands when exploring patterns of sexuality and reproduction.

However, no data could be obtained on health maintenance and perception, sleep and rest, self-perception and self-concept, roles and relationships, coping mechanisms and stress tolerance, or values and beliefs due to sedation, analgesia and invasive mechanical ventilation, as well as a lack of secondary information from family members.

Results

Through the PFS nursing assessment, nursing diagnoses were identified based on the actual or potential problems presented by the patient. Based on the data obtained from nutritional, metabolic, activity, exercise, and elimination patterns, the nursing diagnosis label: 00421 Inadequate fluid volume was primarily established as a priority for restoring the patient's hemodynamic status, (Table 1).



Table 1. Care plan for the nursing diagnosis: 00421 Inadequate fluid volume

Nursing Diagnosis (NANDA)	Nursing Outcome (NOC)
00421. Inadequate fluid volume, which is related to difficulty obtaining fluids and inadequate fluid intake, manifested by altered skin turgor, decreased blood pressure, decreased pulse volume, decreased urine output and increased heart rate.	421. Shock severity: septic. Domain: 2. Physiological health. Class: E. Cardiopulmonary. Indicators: 042101. Decreased systolic blood pressure. 042102. Decreased diastolic blood pressure. 042103. Increased heart rate. 042106. Increased respiratory rate. 042110. Decreased arterial oxygen. 042111. Increased body temperature. 042116. Pallor. 042117. Intravascular coagulation. 042118. Decreased urine output. 042119. Hypoactive bowel sounds.
Nursing Interventions (NIC) 4250. Shock management. Field: 2. Physiological: Complex. Class: Tissue perfusion control. Activities: Monitor vital signs, blood pressure, mental status and urine output. Place the patient in a position that optimizes perfusion. Ensure airway patency. Monitor pulse oximetry. Administer oxygen and/or mechanical ventilation. Use an arterial line to improve the accuracy of blood pressure readings. Perform an arterial blood gas analysis. Monitor trends in hemodynamic parameters. Monitor the factors that determine tissue oxygen supply. Observe for signs of respiratory failure. Monitor laboratory values. Administer crystalloid or colloid fluids intravenously as required. Monitor the hyperdynamic state of septic shock following fluid resuscitation. Antibiotic treatment should be started early and its effectiveness carefully monitored. Monitor blood glucose levels and treat abnormal levels on a case-by-case basis. Monitor hydration status. Monitor kidney function. Administer diuretics. Administer prophylaxis for deep vein thrombosis (DVT) and stress ulcers.	
4255. Shock management: Septic. Field: 2. Physiological: Complex. Tissue perfusion control. Activities: Use the appropriate screening tools (qSOFA and SOFA) once the risk level has been established. Assess vital signs and laboratory values according to the criteria prescribed in the screening tools. Remove any possible sources of infection. Follow the sepsis care package protocol for all individuals with positive screening results and confirmed infections within the established timeframes. Lactate levels should be continuously monitored when they exceed 2 mmol/L. Administer antibiotics promptly. Use vasopressors to maintain a mean arterial pressure (MAP) of at least 65 mmHg. Reassess volume status and tissue perfusion after the initial administration of fluids if hypotension persists (i.e. if MAP is less than 65 mmHg) or if the initial lactate level was equal to or greater than 4 mmol/L.	



Keep a record of all reassessments and ongoing resuscitation measures.
 Eligible at-risk individuals should undergo continuous screening using the Sequential Organ Failure Assessment (SOFA) criteria.
 Monitor early detection indicators.
 Provide instructions on recognizing the signs of infection, taking temperature readings, completing all prescribed courses of antibiotics and the importance of time in the event of serious infections.

Nursing Evaluation

Each indicator was evaluated using the NOC Likert scale based on specific activities performed. Initially, indicators 042101 and 042102 (referring to blood pressure) increased substantially from 2 to 4 with intravenous vasopressor infusion and fluid resuscitation support. The patient's vital signs were constantly monitored, and changes were observed in respiratory rate (indicator 042106) and arterial oxygen saturation (indicator 042110). Invasive mechanical ventilation support increased these indicators from 1 to 3.

Hydration status and renal function (indicators 042116 and 042118) were monitored by calculating hourly urine output. The administration of fluids and diuretics stimulated urine output, improving the indicators from 3 to 5 (from 0.3 to 0.7 ml/kg/h).

Source: Self-developed

Secondly, nursing diagnosis: 00030 Impaired gas exchange, as a priority human response to invasive ventilatory support was established when assessing the patient's respiratory function in the activity-exercise pattern, (Table 2).

Table 2. Care plan for the nursing diagnosis 00030 Impaired gas exchange

Nursing Diagnosis (NANDA)	Nursing Outcome (NOC)
00030. Impaired gas exchange, which is related to an ineffective breathing pattern, manifested by hypercapnia (82 mmHg), abnormal arterial pH (7.23), abnormal skin color, hypoxemia, hypoxia, tachycardia, and tachypnea.	0411. Mechanical ventilation response: Adult. Domain: II. Physiological health. Class: E. Cardiopulmonary. Indicators: 0D41102. Respiratory rate. 041103. Respiratory rhythm. 041107. Vital capacity. 041108. Fraction of inspired oxygen (FiO2) meets oxygen demand. 041109. Partial pressure of oxygen in arterial blood (PaO2). 041110. Partial pressure of carbon dioxide in arterial blood (PaCO2). 041111. Arterial pH. 041112. Oxygen saturation. 041113. Peripheral tissue perfusion. 041124. Difficulty breathing with ventilator. 041127. Adventitious breath sounds. 041134. Atelectasis. 041130. Hypoxia. 041131. Pulmonary infection. 041132. Respiratory secretions. 041137. Cognition. 041128. Restlessness.
Nursing Interventions (NIC)	
3300. Management of mechanical ventilation: invasive.	
Field: 2. Physiological: Complex.	
Class: K. Respiratory Management.	



Activities:

Consult with other healthcare professionals to select the ventilator mode.
Ensure that the ventilator alarms are connected.
Regularly check all ventilator connections.
Administer appropriate muscle relaxants, sedatives and narcotic analgesics, and manage them effectively.
Monitor symptoms indicating increased respiratory effort.
Monitor the patient's physiological and psychological status to assess the effectiveness of mechanical ventilation.
Provide care to relieve the patient's discomfort.
Empty the condensed water from the collectors.
Perform suctioning if there are adventitious sounds and/or increased inspiratory pressures.
Monitor the amount, color and consistency of pulmonary secretions, documenting the results periodically.
Encourage routine assessments to establish weaning criteria.
Establish oral care routine using soft, moist gauze, antiseptic and gentle suction.
Monitor the effects of ventilator changes on oxygenation.

2260. Sedation management.

Domain: 2. Physiological: complex.

Class: H. Medication management.

Activities:

Check for drug allergies.
Review medications and check for contraindications to sedation.
Review the written informed consent.
Assess the patient's level of consciousness and protective reflexes before proceeding with sedation.
Determine baseline vital signs, oxygen saturation, ECG, height, and weight.
Ensure that emergency resuscitation equipment was readily available: a source for administering 100 % O₂, emergency medication, and a defibrillator.
Administer medication according to the physician's prescription, carefully adjusting the dosage according to the patient's response.
Monitor the patient's level of consciousness, airways, vital signs, oxygen saturation, end-tidal CO₂ and ECG during the procedure.
Monitor for any adverse effects resulting from the medication after it has been administered.
Use a sedation scale (RASS) to determine the level of consciousness.
Ensure antagonists are available and administer them as appropriate according to the physician's prescription.
Keep a record of actions taken and responses observed.

Nursing Evaluation

The results of the assessed NOCs improved substantially. This included improvements in respiratory rate, rhythm and capacity, which increased from level 1 to level 4 thanks to specific changes made to the mechanical ventilator and an increase in the sedation dose as medically indicated. The response and consciousness status were constantly assessed using the Richmond Agitation and Sedation Scale, with indicator 041128 improving from 2 to 4. Arterial pH and carbon dioxide levels remained at 3 as there was no improvement in the signs of respiratory acidosis.

Source: Self-developed

The nursing diagnosis 00031 Ineffective airway clearance, was also established in response to a decreased neurological response due to induced sedation and analgesia, as well as increased secretion production due to irritation from the orotracheal tube, (Table 3).



Table 3. Care plan for the nursing diagnosis 00031 Ineffective airway clearance

Nursing Diagnosis (NANDA) 00031. Ineffective airway clearance, which is related to secretion retention, manifested by adventitious breath sounds, altered breathing pattern, and tachypnea.	Nursing Outcome (NOC) 04:10. Respiratory status: Airway patency. Domain: II. Physiological health. Class: E. Cardiopulmonary. Indicators: 041004. Respiratory rate. 041005. Respiratory rhythm. 041012. Ability to clear secretions. 04:10:22. Body Mass Index (BMI). 041007. Abnormal breath sounds. 041018. Use of accessory muscles. 041028. Secretions in the airways. 041030. Inflammation of airway tissues.
Nursing Interventions (NIC) 3900. Airway suctioning. Field: 2. Physiological: Complex. Class: K. Respiratory management. Activities: Wash hands. Use universal precautions. Use personal protective equipment. Determine the need for oral and/or tracheal suctioning. Auscultate breath sounds before and after suctioning. Provide sedation. Hyperoxygenate with 100% oxygen for at least 30 seconds using the ventilator. Use closed-system suction. Use sterile disposable equipment for each suctioning procedure. If a closed tracheal suction system or oxygen insufflation device adapter is used, leave the patient connected to the ventilator during suctioning. Monitor the patient's oxygenation status. Stop tracheal suctioning and administer supplemental oxygen if the patient experiences bradycardia, increased ventricular extrasystoles and/or desaturation. Observe and check the color, amount and consistency of secretions.	
Nursing Evaluation Using the established NIC, it was possible to increase the Likert score for certain NOC indicators. The patient's respiratory rate decreased to normal values, i.e., the score for indicator 041004 increased from 3 to 5, and indicator 041005 improved significantly. In addition, rales, which are pathological breathing sounds, disappeared completely.	
Source: Self-developed	

Regarding the findings in the metabolic nutritional pattern, nursing diagnosis 00044 Impaired tissue integrity, was established, (Table 4).



Table 4. Care plan for the nursing diagnosis 00044 Impaired tissue integrity

Nursing Diagnosis (NANDA)	Nursing Outcome (NOC)
00044. Impairment of tissue integrity, which is related to fluid and excretion imbalance, manifested by abnormal tissue growth, skin color changes, and persistent erythema.	1103. Wound healing: Second intention. Domain: II. Physiological health. Class: L. Tissue integrity. Indicators: 110301. Granulation. 110320 Scar formation. 110321. Decrease in wound size. 110303. Purulent discharge. 110304. Serous discharge. 110305. Bloody discharge. 110307. Erythema of surrounding skin. 110322. Wound inflammation. 110308. Perilesional edema. 110317. Foul odor from the wound.
Nursing Interventions (NIC) 3660. Wound care. Field: 2. Physiological: Complex. Class: L. Skin/wound management. Activities: Monitor oxygenation status before and after changing position. Place the patient in the specified therapeutic position. Ensure correct body alignment. Elevate the affected body part. Place supports on edematous areas (upper and lower limbs). Avoid placing the patient in positions that increase pain. Minimize friction and shear forces when repositioning the patient. Turn the patient as a whole. Nursing Evaluation The NIC applied to increase the NOC score was slightly effective, as the surgical wound presented severe alterations. The NOC indicators increased from 1 to 2 with wound healing, and secretions and foul odor decreased slightly. However, the score remained unchanged in terms of granulation, reduction in size, and perilesional edema.	
Source: Self-developed	

Discussion

The patient was in septic shock and presented with severe hypotension despite resuscitation with fluid therapy. The patient also required a vasopressor infusion to maintain a mean arterial pressure of over 65 mmHg and had hyperlactacidaemia of over 20 mg/dL despite receiving adequate fluids. This required specific interventions to address the nursing diagnosis (00421) inadequate fluid volume, as well as to assess the severity of the patient's shock. The authors ^(13,14) agree that administering vasoactive agents and fluid resuscitation in cases of sepsis is important for improving tissue perfusion and achieving different objectives.



Similarly, one researcher emphasizes the vital importance of nursing professionals identifying septic shock, given that it causes changes that endanger the patient's health. The author describes various nursing interventions, such as administering intravenous fluids, medications and oxygen therapy, as well as monitoring fluid balance, to improve the patient's health and prevent future complications ⁽¹⁵⁾.

On the other hand, for the management of nursing diagnosis (00030) Impaired gas exchange, invasive mechanical ventilation was used with volumes between 6 and 12 ml/kg for patients with acute respiratory distress. For the management of nursing diagnosis (00031) Ineffective airway clearance, the recommendation to improve respiratory ventilation is to suction airway secretions, and to limit the risk of bronchoaspiration, raise the head of the bed between 30° and 45° ⁽¹⁶⁾. These interventions are related to what has been described by various authors ^(17,18), who have shown that activities such as secretion removal facilitate ventilatory and blood oxygenation mechanisms and help prevent complications, mainly pneumonia caused by microorganisms.

The patient's septic process originated from a contaminated surgical wound. This required the development of interventions to update the management of chronic wounds using different healing methods, which is considered of great importance to the nursing profession. This posed a significant challenge for the nursing team, as it would be difficult to heal the affected area without the appropriate treatments. There was also a possibility of local and systemic complications developing ⁽¹⁹⁾.

Various interventions are available, such as advanced wound care, which promotes healing through the principle of moist wound healing. However, passive products are considered a negative practice in the management of chronic wounds as they have a low absorption capacity and can cause infections ⁽²⁰⁾.



Among the limitations of the study are the patient's health status and restrictions on family visits, which meant that subjective information could not be obtained.

Conclusions

Applying the NP to a patient with a medical diagnosis of intestinal septic shock enabled us to identify dysfunctional health patterns and formulate six nursing diagnoses. These were then ranked according to the AREA model. Expected health outcomes were selected and interventions to address these issues were presented. These interventions resulted in moderate improvements in hemodynamic, respiratory and urinary functions, demonstrating that specialized care for critically ill patients can address actual and/or potential health issues. The patient's health status improved slightly during the application of the nursing process. However, there was evidence of a high mortality risk.

On the other hand, among the updates to the standardized language of the nursing profession is an innovation in the selection of NOCs that specifies the type of shock severity to be assessed after an NIC. Examples include Shock severity: septic and Shock severity: cardiogenic. Similarly, when selecting an NIC, the type of shock to which the activities were applied is specified, such as septic shock management or cardiac shock management. These innovations in nursing science highlight the need to specify the type of NIC and define the activities aimed at improving patients' health status.

For nursing professionals, designing and applying the NP is a very important part of emergency, ICU and hospital services. This helps them to clearly identify human responses and pathophysiological processes, as well as the appropriate treatment, thereby improving survival rates for these patients.

Conflict of interest

The authors stated that they have no conflicts of interest.



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Artificial intelligence

The authors declare that they have not used any artificial intelligence resources in any of the sections of this manuscript.

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